

# **SBIAHU CSE 114 Fall 2025**

## **Midterm 1 Practice Questions**

**Midterm is scheduled on: Sunday, 11/16/2025 from 10:20 to 12:20.**

- It is an in-person, paper based test, you don't need any computer/laptop for the test.
- **Please be on time; students arriving more than 10 minutes late will NOT be admitted to the exam.**
- The exam is CLOSED-BOOK, CLOSED-NOTES. No electronic or manual assistive devices (other than a writing implement) may be used during the exam, so leave laptops, tablets, smartphones, calculators, smartwatch, slide rules, etc., at home (or, at least, turn them off and stow them in your backpacks so that they are out of reach and cannot be seen or consulted during the exam).
- Final answers should be written using (ideally) non-erasable blue or black ink. DO NOT use pink, red, or purple ink under any circumstances!

- **Midterm 1 Content:**

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- Elementary programming
- Selections
- Number Systems
- Mathematical Functions, Characters and Strings
- Loops
- Methods
- Array

First 6-7 questions of the test have sub-parts which will be short to medium size questions. These questions are TRUE/FALSE, multiple choice, fill in the blanks, rewrite the code segment, code analysis (what will be the output of this program) types.

In the last 3-4 problems (questions) you have to write the complete programs. In these questions you must properly have the import statement (if required), must define the class properly, must have correct main method header, should have proper variable declaration and initialization, proper statement termination, etc.

**Do not miss any exams!** Make-up exams will be given only in extenuating circumstances at the discretion of the instructor, subject to the university policies. Students who miss an exam for a valid reason must contact the instructor immediately to schedule a make-up exam at the earliest possible time.

**Information on this page will be provided.**

**The string class:**

- `charAt(n)`: returns the character at index n. indices are numbered from 0- (length - 1)
- `length()`: returns the number of characters contained in the current String
- `toLowerCase()`: returns all-lowercase version of the current String
- `toUpperCase()`: returns all-uppercase version of the current String
- `substring(i, j)`: returns a new String containing the characters from index i up to but not including, index j
- `substring(i)`: returns a new String containing the characters from index i up to the end of the String.
- `indexOf(s)`: returns the index of the first occurrence of the String s in the current String, or -1 if it cannot be found.
- `equals(s)`: returns true if its argument matches the current String exactly, including case.(compare to the `equalsIgnoreCase()` method)

**The Math class:**

- `abs(x)`: returns the absolute value of x.
- `max(x, y)` : returns the larger of x and y.
- `min(x, y)`: returns the minimum of x and y.
- `pow(b, e)`: raises b to the e-th power.
- `random()`: returns a random double from 0.0 up to (not including) 1.
- `round(x)`: rounds x to the nearest integer value.
- `sqrt(x)`: returns the square root of x, where  $x \geq 0$ .

1. Solve the following questions on number systems

[5 + 5]

- a. Convert the following number in hexadecimal representation to decimal representation. Show your work for partial credit.  
10AF1
- b. Convert the following decimal number into binary representation. Show your work for partial credit.  
235

2. What will be the output of the following programs?

[5 + 5]

a.

```
public class Problem {  
    public static void main(String[] args) {  
        int a = 6;  
        int b = a++;  
        System.out.println(a);  
        System.out.println(b);  
        a = 6;  
        b = ++a;  
        System.out.println(a);  
        System.out.println(b);  
    }  
}
```

b.

```
public class Problem {  
    public static void main(String[] args) {  
        double x = 1.0;  
        double y = 5.0;  
        double z = x-- + (++y);  
        System.out.println(x);  
        System.out.println(y);  
        System.out.println(z);  
    }  
}
```

3. What will be the output of the following programs?

[5 + 5]

a.

```
public class Problem {
    public static void main(String[] args) {
        int x = 10;
        int y = 4;
        if(x < 5 && ++y == 5)
            System.out.println("x is : " + x + " and, y is: " + y);
        else
            System.out.println("x is : " + x + " and, y is: " + y);
    }
}
```

b.

```
public class Problem {
    public static void main(String[] args) {
        int x = 10;
        int y = 4;
        if(x < 5 || ++y == 5)
            System.out.println("x is : " + x + " and, y is: " + y);
        else
            System.out.println("x is : " + x + " and, y is: " + y);
    }
}
```

c.

```
public class Problem {
    public static void main(String[] args) {
        int x = 10;
        int y = 4;
        if(++y == 6 || x < 11 )
            System.out.println("x is : " + x + " and, y is: " + y);
        else
            System.out.println("x is : " + x + " and, y is: " + y);
    }
}
```

4. What will be the output of the following programs?

[5 + 5]

a.

```
public class Problem {  
    public static void main(String[] args) {  
        boolean x = 1 + 4 * 4 < 5 * (4 - 3) - 1 && 4 - 3 < 5;  
        System.out.println(x);  
    }  
}
```

b. Show the output of the following program: when the user provides the following inputs:

input an integer in the range 1 to 4 : 1  
input an integer in the range 1 to 4 : 2  
input an integer in the range 1 to 4 : 3  
input an integer in the range 1 to 4 : -10

```
import java.util.Scanner;  
public class Simple {  
    public static void main(String[] args) {  
        System.out.println("input an integer in the range 1 to 4");  
        Scanner stdin = new Scanner(System.in);  
        int intInput = stdin.nextInt();  
        switch(intInput) {  
            case 1:  
                System.out.println("you entered ONE");  
                break;  
            case 2:  
                System.out.println("you entered TWO");  
            case 3:  
                System.out.println("you entered THREE");  
                break;  
            default:  
                System.out.println("your input out of range");  
        }  
        stdin.close();  
    }  
}
```

5. Answer the following questions:

[5 + 5]

a. Rewrite the following if statement using the conditional operator

```
if(ages >= 16)
    ticketPrice = 20;
else
    ticketPrice = 10;
```

b. What is y after the following switch statement is executed? Rewrite the code using an if-else statement.

```
int x = 3, y = 3;
switch(x + 3) {
case 6:
    y = 1;
default:
    y += 1;
}
```

6. Answer the following questions:

[5 + 5]

a. Rewrite the following program using switch statement.

```
import java.util.Scanner;
public class MyClass {
    public static void main(String[] args) {
        Scanner stdin = new Scanner(System.in);
        System.out.println("enter an integer: 1, 2, 3 or 4");
        int x = stdin.nextInt();
        if(x == 1)
            System.out.println("Freshman");
        else if(x == 2)
            System.out.println("Sophomore");
        else if(x == 3)
            System.out.println("Junior");
        else if(x == 4)
            System.out.println("Senior");
        else
            System.out.println("Improper input");
        stdin.close();
    }
}
```

b. Show the output of the following statements (write a program to verify your results):

- i) System.out.println("1" + 1);
- ii) System.out.println('1' + 1);
- iii) System.out.println("1" + 1 + 1);
- iv) System.out.println("1" + (1 + 1));
- v) System.out.println('1' + 1 + 1);

7. Write down a program to find out the greatest common divisor for two integers. Program should prompt the user to input two integers and output the greatest common divisor. Here are some example runs: [10]

Enter the first integer: 12  
Enter the second integer: 16  
The greatest common divisor is: 4

Enter the first integer: 3  
Enter the second integer: 5  
The greatest common divisor is: 1

8. Write a program that simulates rolling of a pair of dice. You can simulate rolling one die by choosing one of the integers 1, 2, 3, 4, 5, or 6 at random. The number you pick represents the number on the die after it is rolled. Your program should report the number showing on each die as well as the total roll. For example: [10]

The first die comes up 3  
The second die comes up 5  
Your total roll is 8

9. Write a program that randomly generates an integer between 1 and 12 (including 1 and 12) and displays the English month names: January, February... December for the numbers 1, 2 ...12 accordingly. [10]

10. Write down a complete program to generate a random string. It should use an integer variable *len* and a character variable *ulCase*. Program should prompt the user to input the values for *len* and *ulCase*. The integer parameter *len* will be used to determine the size of the random string whereas, the parameter *ulCase* will determine whether the generated string will be in uppercase or in lowercase. If *ulCase* = 'u' or 'U' generate random uppercase string. If *ulCase* = 'l' or 'L' generate a lowercase random string. [10]

Here are some example run:  
Enter the length of string: 8  
Enter the case input : L  
Random string : mwmwatvv

Enter the length of string: 5  
Enter the case input : u  
Random string : IYOGA